Embedded Systems Homework #5

Nidhi Borhade
StudentId : 109998404

October 2023

Contents

1 Configure Disk Quotas

1. Download Quota

```
    sudo apt update
    sudo apt install quota
```

2. Prepare File System

```
    cd etc/
    sudo nano fstab
    > Enter Quota Cofiguration
    /dev/sdX        /mnt/data        ext4        defaults,usrquota,grpquota    0 0
```

3. Activate Quotas

```
        sudo quotacheck -cug /mnt/data
```

4. Initalize User

```
        sudo edquota -u Felix
```

5. Start Quota

```
        sudo quotaon /mnt/data
```

6. Check Quotas

```
        quota -u Felix
```

2 Mail Web-Server

1. Allocate Disk Space:

   When setting up your server or virtual machine, allocate sufficient disk space to meet your overall storage requirements, including the 500 MB for email and 200 MB for web. Install and Configure Email Server:

2. Install your preferred email server software (e.g., Postfix, Dovecot). Implement user quotas on the email storage directory to restrict the mailbox size to 500 MB per user. The specific steps will depend on your chosen email server software. Install and Configure Web Server:

3. Install a web server software (e.g., Apache, Nginx). Configure virtual hosts for your websites and set up the document root to allocate the desired 200 MB of space.

4. User Management for Personal Web Space:

   If you plan to offer personal web space services, create user accounts for individuals who need web hosting. Set user quotas to limit each user's web space to 200 MB. Quota Management:

5. Implement quotas using tools like quota and edquota on your Linux system to set size limits for users. Refer to your specific Linux distribution's documentation for detailed instructions. Regular Monitoring:

6. Regularly monitor disk usage to ensure that users and services do not exceed their allocated quotas. You can use tools like quota and system monitoring utilities.

7. Backup and Data Management:

   Implement a backup strategy to protect your email and web data. Ensure that your backup processes account for quotas and include email and web data in your backups.

3 Backups

1. tar *, preserves the user and group ownership of the files and directories in the archive by default*

```
    tar -cvf backupHomeWork.tar homework/

    1) c Flag - creates a new Archiv
    2) v Flag - verbose flag, will print a
       list of the archived data
    3) f Flag - file Flag, gives the option to
        give a name to the archive in this
       example "backupHomeWork.tar"
    4) other option is to use z flag too, it
       will compress the archive using gzip
```

2. gzip/gunzip *Pay attention, gzip by itself does not safe safe user, group or file permission, its only a tool to safe storage*
   So if you want to use it to safe all the meta information too, tar the files first and then gzip them.

```
    tar -cvf backupHomeWork.tar homework/
    gzip backUpHomework.tar

    OR
    tar -czf backUpHomeWork.tar.bz2 homework/

      1) z Flag - signalises tar to use gzip
```

3. bzip2 *Same goes with bzip2*

```
     tar -cvf backupHomeWork.tar homework/
     bzip2 backUpHomework.tar

    OR
     tar -cjf backUpHomeWork.tar.bz2 homework/

     1) j Flag - signalises tar to use bzip2
```